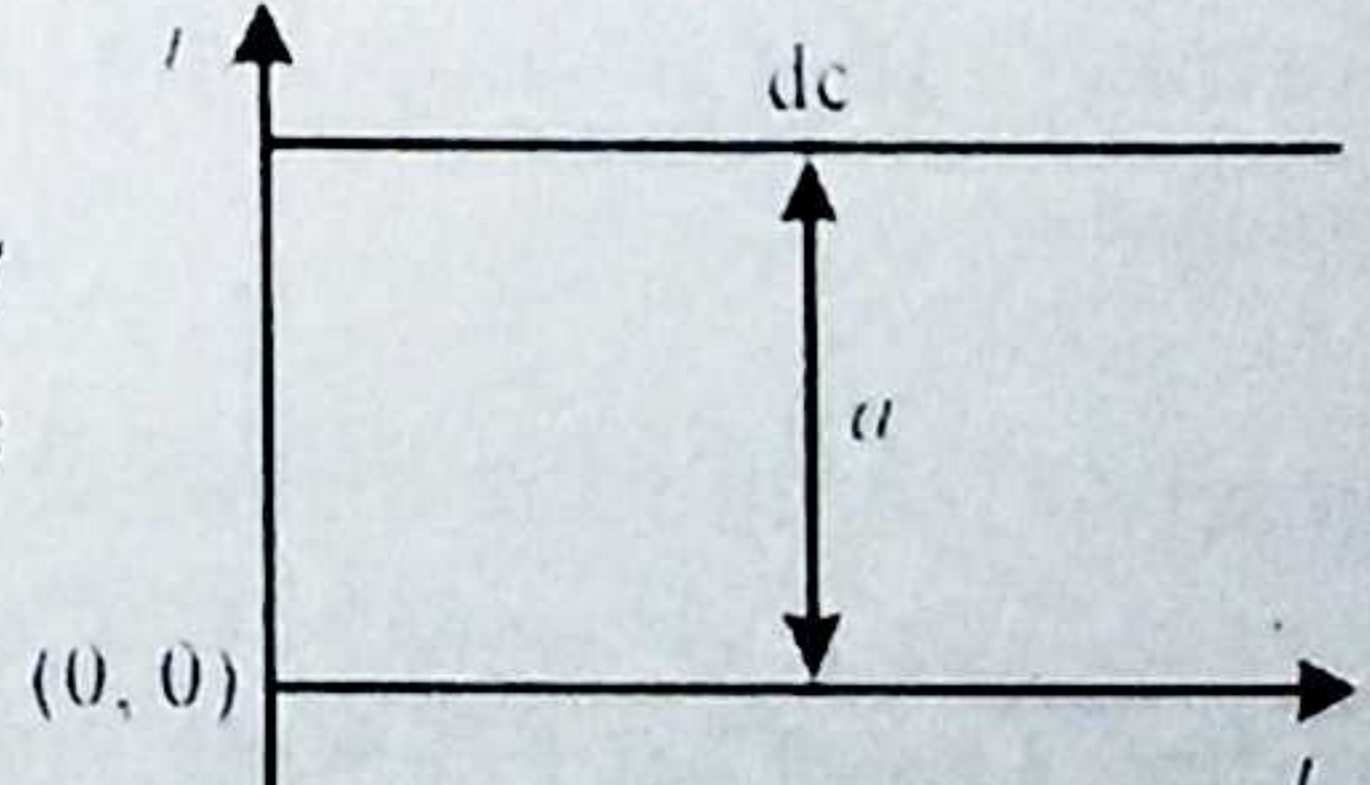


Alternating Current

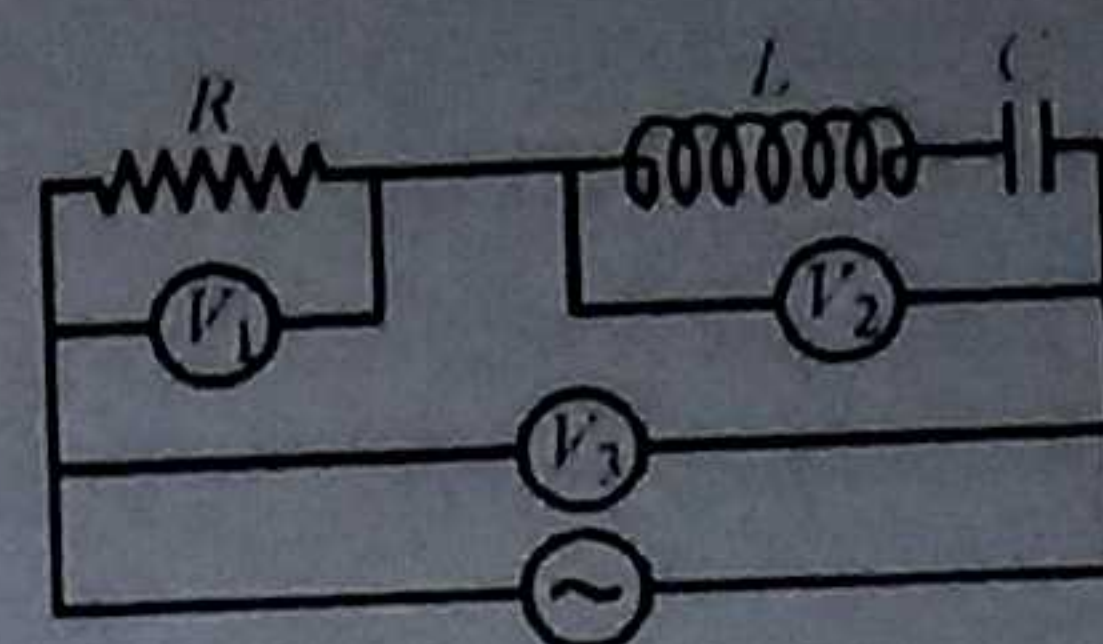
Type of CPP - Objective & Subjective

Level - MODERATE

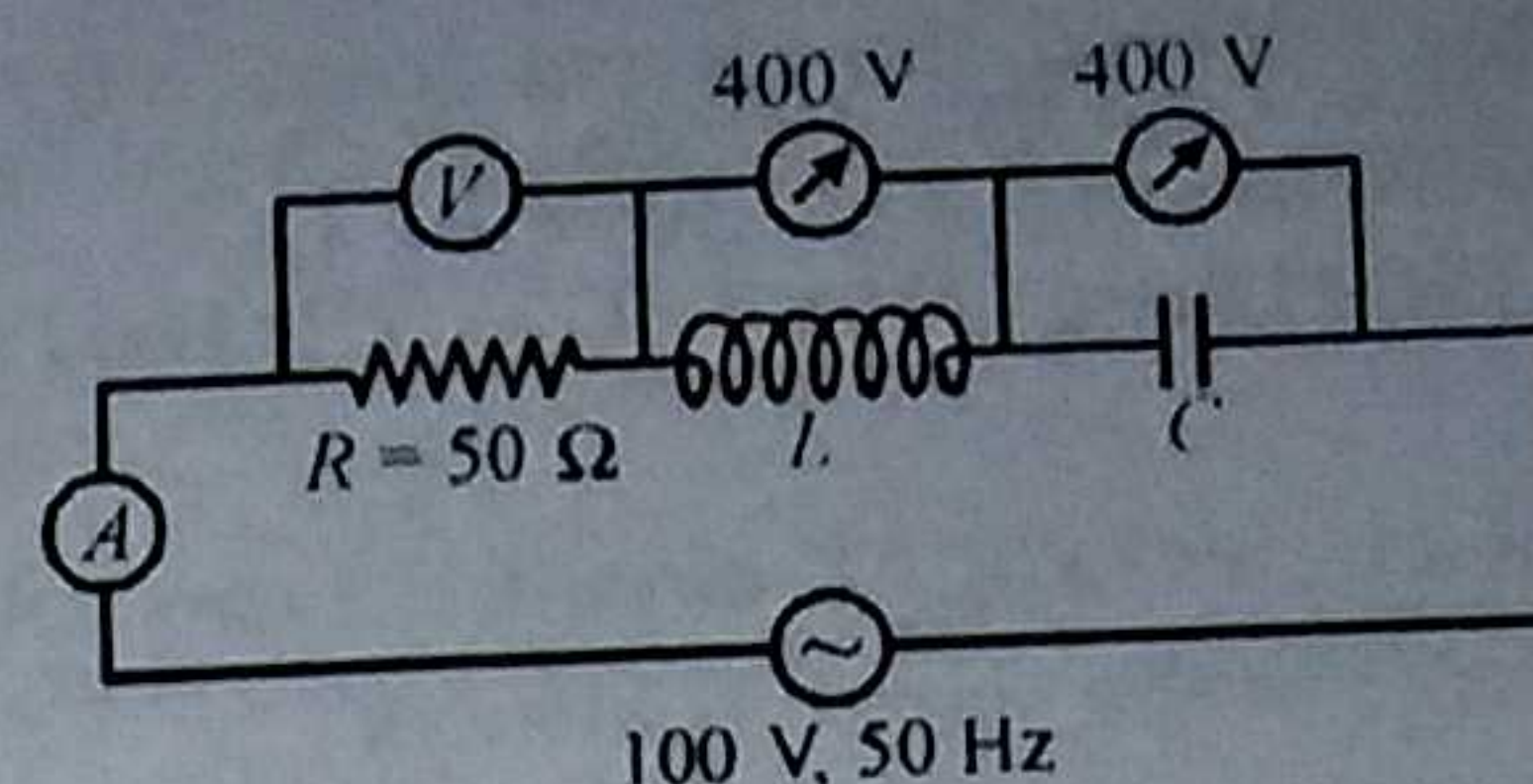
1. In an LCR circuit, the capacitance is made one-fourth, when in resonance. Then what should be the change in inductance, so that the circuit remains in resonance ?
 (A) 4 times (B) 1/4 times (C) 8 times (D) 2 times
2. A 0.21-H inductor and a 88-Ω resistor are connected in series to a 220-V, 50-Hz AC source. The current in the circuit and the phase angle between the current and the source voltage are respectively. (Use $\pi = 22/7$)
 (A) 2A, $\tan^{-1} 3/4$ (B) 14.4A, $\tan^{-1} 7/8$ (C) 14.4A, $\tan^{-1} 8/7$ (D) 3.28A, $\tan^{-1} 2/11$
3. An LCR series circuit with 100 Ω resistance is connected to an AC source of 200 V and angular frequency 300 radians per second. When only the capacitance is removed, the current lags behind the voltage by 60°. When only the inductance is removed, the current leads the voltage by 60°. Then the current and power dissipated in LCR circuit respectively :
 (A) 1A, 200 watt (B) 1A, 400 watt (C) 2A, 200 watt (D) 2A, 400 watt
4. Power factor may be equal to 1 for :
 (A) pure inductor (B) pure capacitor (C) pure resistor (D) An LCR circuit
5. The household supply of electricity is at 220 V rms value and 50 Hz. Find the peak voltage and the least possible time in which the voltage can change from the rms value to zero.
6. The average power delivered to a series AC circuit is given by (symbols have their usual meaning) :
 (A) $E_{\text{rms}} I_{\text{rms}}$ (B) $E_{\text{rms}} I_{\text{rms}} \cos \phi$ (C) $E_{\text{rms}} I_{\text{rms}} \sin \phi$ (D) zero
7. The current in a discharging LR circuit is given by $i = I_0 e^{-t/\tau}$ where τ is the time constant of the circuit calculate the rms current for the period $t = 0$ to $t = \tau$.
8. Energy dissipates in LCR circuit in :
 (A) L only (B) C only (C) R only (D) All of these
9. The potential difference V across and the current I flowing through an instrument in an AC circuit are given by
 $V = 5 \cos \omega t$ volt
 $I = 2 \sin \omega t$ volt
 The power dissipated in the instrument is :
 (A) Zero (B) 5 watt (C) 10 watt (D) 2.5 watt
10. If a direct current of value 'a' ampere is superimposed on an alternating current $I = b \sin \omega t$ flowing through a wire. What is the effective (rms) value of the resulting current in the circuit ?

11. Find the average for saw-tooth voltage of peak value V_0 from $t = 0$ to $t = 2T$ as shown in figure.
12. The value of power factor $\cos \phi$ in series LCR circuit at resonance is :
 (A) zero (B) 1 (C) 1/2 (D) 1/2 ohm

13. A series LCR circuit containing a resistance of 120 ohm has angular resonance frequency $4 \times 10^3 \text{ rad s}^{-1}$. At resonance, the voltage across resistance and inductance are 60V and 40V respectively. The values of L and C are respectively :
 (A) 20 mH, $25/8 \mu\text{F}$ (B) 2 mH, $1/35 \mu\text{F}$ (C) 20 mH, $1/40 \mu\text{F}$ (D) 2 mH, $25/8 \mu\text{F}$
14. An electric bulb is designed to operate at 12 volts DC. If this bulb is connected to an AC source and gives same brightness. What would be the peak voltage of the source ?
15. A coil has an inductance of $\frac{2.2}{\pi} \text{ H}$ and is joined in series with a resistance of 220Ω . When an alternating e.m.f. of 220 V at 50 cps is applied to it, then the wattless component of the rms current in the circuit is resistance. The ratio of heat produced in the two resistances in the same time interval will be :
 (A) 5 ampere (B) 0.5 ampere (C) 0.7 ampere (D) 7 ampere
16. A direct current of 2A and an alternating current having a maximum value of 2A flow through two identical resistances. The ratio of heat produced in the two resistances in the same time interval will be :
 (A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 4 : 1
17. A sinusoidal AC current flows through a resistor of resistance R . If the peak current is I_p , then average power dissipated is :
 (A) $I_p^2 R \cos \theta$ (B) $\frac{1}{2} I_p^2 R$ (C) $\frac{4}{\pi} I_p^2 R$ (D) $\frac{1}{\pi^2} I_p^2 R$
18. What is the rms value of an alternating current which when passed through a resistor produces heat, which is thrice that produced by a current of 2 ampere in the same resistor in the same time interval ?
 (A) 6 ampere (B) 2 ampere (C) $2\sqrt{3}$ ampere (D) 0.65 ampere
19. A resistor and a capacitor are connected to an AC supply of 200 volt, 50 Hz in series. The current in the circuit is 2 ampere. If the power consumed in the circuit is 100 watt, then the resistance in the circuit is :
 (A) 100Ω (B) 25Ω (C) $\sqrt{125 \times 75} \Omega$ (D) 400Ω
20. The impedance of a series circuit consists of 3 ohm resistance and 4 ohm reactance. The power factor of the circuit is :
 (A) 0.4 (B) 0.6 (C) 0.8 (D) 1.0
21. An electric bulb and a capacitor are connected in series with an AC source. On increasing the frequency of the source, the brightness of the bulb :
 (A) increase (B) decreases
 (C) remains unchanged (D) sometimes increases and sometimes decreases
22. By what percentage the impedance in an AC series circuit should be increased so that the power factor changes from $(1/2)$ to $(1/4)$ (when R is constant) ?
 (A) 200% (B) 100% (C) 50% (D) 400%
23. Average power consumed in an AC series circuit is given by (symbols have their usual meaning) :
 (A) $E_{\text{rms}} I_{\text{rms}} \cos \phi$ (B) $(I_{\text{rms}})^2 R$ (C) $\frac{E_{\text{max}}^2 R}{2(|z|)^2}$ (D) $\frac{I_{\text{max}}^2 |z| \cos \phi}{2}$
24. A resistor of resistance 100Ω is connected to an AC source $\epsilon = (12 \text{ V}) \sin (250 \pi \text{ s}^{-1})t$. Find the power consumed by the bulb.
25. In an AC circuit the instantaneous values of current and applied voltage are respectively $i = 2 (\text{Amp}) \sin (250 \pi \text{ s}^{-1})t$ and $\epsilon = (10 \text{ V}) \sin \left((250 \pi \text{ s}^{-1})t + \frac{\pi}{3} \right)$. Find the instantaneous power drawn from the source at $t = \frac{2}{3} \text{ ms}$ and its average value.
26. A resistor R , an inductor L and a capacitor C are connected in series to an oscillator of frequency n . If the resonant frequency is n_r , then the current lags behind voltage, when :
 (A) $n = 0$ (B) $n < n_r$ (C) $n = n_r$ (D) $n > n_r$

27. A resistor R , and inductor L , a capacitor C and voltmeters V_1 , V_2 and V_3 are connected to an oscillator in the circuit as shown in the adjoining diagram. When the frequency of the oscillator is increased, upto resonance frequency, the voltmeter reading (at resonance frequency) is zero in the case of :
- (A) voltmeter V_1 (B) voltmeter V_2
(C) voltmeter V_3 (D) All the three voltmeters



28. In the series LCR circuit, the voltmeter and ammeter readings are :
- (A) $V = 100$ volt, $I = 2$ amp
(B) $V = 100$ volt, $I = 5$ amp
(C) $V = 1000$ volt, $I = 2$ amp
(D) $V = 300$ volt, $I = 1$ amp



29. The dielectric strength of air is 3.0×10^6 V/m. A parallel plate air capacitor has area 20 cm^2 and plate separation $\sqrt{2} \text{ mm}$. Find the maximum rms voltage of an AC source which can be safely connected to this capacitor.
30. A 100 volt AC source of angular frequency 500 rad/s is connected to a LCR circuit with $L = 0.8 \text{ H}$, $C = 5 \mu\text{F}$ and $R = 10 \Omega$. all connected in series. The potential difference across the resistance is :
- (A) $\frac{100}{\sqrt{2}}$ volt (B) 100 volt (C) 50 volt (D) $50\sqrt{3}$ volt

31. In an AC series circuit, the instantaneous current is zero when the instantaneous source voltage is maximum. Connected to the source may be a :
- (A) pure inductor (B) pure capacitor
(C) pure resistor (D) combination of an inductor and a capacitor

32. A pure resistive circuit element X when connected to an AC supply of peak voltage 200 V gives a peak current of 5 A which is in phase with the voltage. A second circuit element Y , when connected to the same AC supply also gives the same value of peak current but the current lags behind by 90° . If the series combination of X and Y is connected to the same supply, what will be the rms value of current ?
- (A) $\frac{10}{\sqrt{2}}$ amp (B) $\frac{5}{\sqrt{2}}$ amp (C) $\frac{5}{2}$ amp (D) 5 amp

33. An electric bulb is designed to consume 55 W when operated at 110 volts . It is connected to a 220 V , 50 Hz line through a choke coil in series. What should be the inductance of the coil for which the bulb gets correct voltage ?

34. In a series LCR circuit with an AC source $R = 330 \Omega$, $C = 20 \mu\text{F}$, $L = 1.0 \text{ henry}$, $E_{\text{rms}} = 50 \text{ V}$ and $\nu = 50/\pi \text{ Hz}$. Find (a) the rms current in the circuit and (b) the rms potential differences across the capacitor, the resistor and the inductor.

35. Consider the situation of the previous problem find the average electric field energy stored in the capacitor and the average magnetic field energy stored in the coil.

36. The peak value of an alternating e.m.f. E given by $E = E_0 \cos \omega t$ is 10 volt and frequency is 50 Hz . At time $t = (1/600) \text{ sec}$, the instantaneous value of e.m.f. is :
- (A) 10 volt (B) $5\sqrt{3}$ volt (C) 5 volt (D) 1 volt

37. The voltage of an AC source varies with time according to the equation, $V = 100 \sin 100 \pi t \cos 100 \pi t$. Where t is in second and V is in volt. Then :

- (A) the peak voltage of the source is 100 volt (B) the peak voltage of source is $(100/\sqrt{2})$ volt
(C) the peak voltage of the source is 50 volt (D) the frequency of the source is 50 Hz

38. An alternating voltage is given by $e = e_1 \sin \omega t + e_2 \cos \omega t$. Then the root mean square value of voltage is given by :

- (A) $\sqrt{e_1^2 + e_2^2}$ (B) $\sqrt{e_1 + e_2}$ (C) $\sqrt{\frac{e_1^2 + e_2^2}{2}}$ (D) $\sqrt{\frac{e_1^2 + e_2^2}{2}}$

39. A 20 volts 5 watt lamp is used on AC mains of 200 volts and $\frac{50}{\pi}\sqrt{11}$ c.p.s. Calculate the (i) capacitor, (ii) inductor, to be put in the series to run the lamp. (iii) How much pure resistance should be included in place of the above device so that the lamp can run on its rated voltage. (iv) Which is more economical, the capacitor, the inductor or the resistor.
40. When 100 V DC is applied across a solenoid, a steady current of 1A flows in it. When 100 V AC is applied across the same solenoid, the current drops to 0.5 amp. If the frequency of the AC source is $150\sqrt{3}/\pi$ Hz, the impedance and inductance of the solenoid are :
 (A) $200\ \Omega$ and $1/3$ H (B) $100\ \Omega$ and $1/16$ H (C) $200\ \Omega$ and 1.0 H (D) $1100\ \Omega$ and $1/117$ H
41. If in a series LCR AC circuit, the rms voltage across L , a capacitor C and voltmeters V_1 , V_2 and V_3 respectively, then the voltage of the source is always :
 (A) equal to $V_1 + V_2 + V_3$ (B) equal to $V_1 - V_2 + V_3$ (C) more than $V_1 + V_2 + V_3$ (D) None of these
42. A circuit has a coil of resistance 50 ohms and inductance $\frac{3}{\pi}$ henry. It is connected in series with a condenser of $\frac{40}{\pi}$ μ F and AC supply voltage of 200 V and 50 cycles/sec. Calculate :
 (i) the impedance of the circuit.
 (ii) the p.d. across inductance coil and condenser.
43. A circuit contains a resistance of 20 ohm and inductance of $\frac{2}{\pi}$ henry and an alternating effective e.m.f. of 500 voltage a frequency of 120 cycles per second applied to it. Find the value of effective current in the circuit and power factor.
44. A coil draws a current of 1.0 ampere and a power of 100 watt from an AC. Source of 100 Volt and $\frac{5\sqrt{22}}{\pi}$ Hertz. Find the inductance and resistance of the coil.
45. An inductor $2/\pi$ Henry, a capacitor $100/\pi$ μ F and a resistor $75\ \Omega$ are connected in series across a source of EMF $V = 10 \sin 100\pi t$. (a) Find the impedance of the circuit. (b) Find the energy dissipated in the circuit in 20 minutes.
46. A series circuit consists of a resistance, inductance and capacitance. The applied voltage and the current at any instant are given by

$$E = 141.4 \cos(5000t - 10^\circ)$$
 and

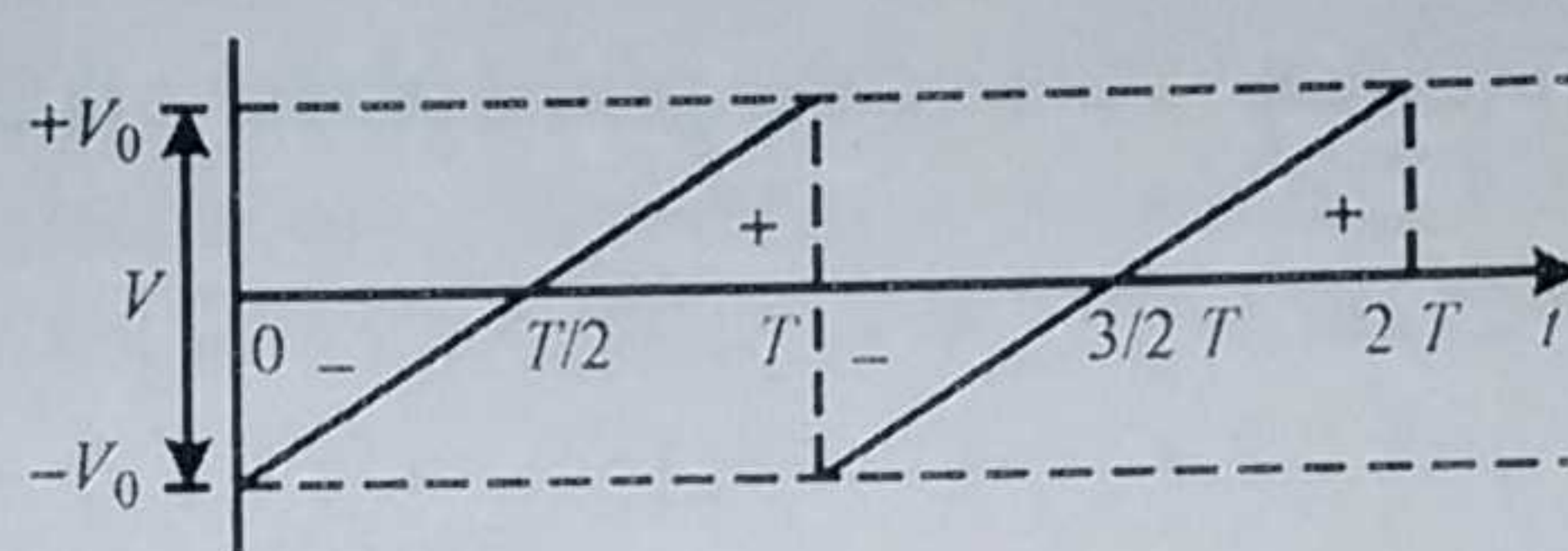
$$I = 5 \cos(5000t - 370^\circ)$$
 The inductance is 0.01 henry. Calculate the value of capacitance and resistance.
47. An inductance of 2.0 H, a capacitance of 18 μ F and a resistance of 10 k Ω are connected to an AC source of 20 V with adjustable frequency. (a) What frequency should be chosen to maximise the current (RMS) in the circuit? (b) What is the value of this maximum current (RMS)?
48. An inductor-coil a capacitor and an AC source of rms voltage 24 V are connected in series. When the frequency of the source is varied a maximum rms current of 6.0 A is observed. If this inductor coil is connected to a battery of emf 12 V and internal resistance 4.0 Ω , what will be the steady current?
49. In an L - R series circuit ($L = \frac{175}{11}$ mH and $R = 12\ \Omega$), a variable emf source ($V = V_0 \sin \omega t$) of $V_{\text{rms}} = 130\sqrt{2}$ V and frequency 50 Hz is applied. The current amplitude in the circuit and phase of current with respect to voltage are respectively. (Use $\pi = 22/7$)
 (A) 14.14A, 30° (B) $10\sqrt{2}$ A, $\tan^{-1} \frac{5}{12}$ (C) 10A, $\tan^{-1} \frac{5}{12}$ (D) 20A, $\tan^{-1} \frac{5}{12}$
50. In an AC circuit, a resistance of R ohm is connected in series with an inductance L . If phase angle between voltage and current be 45° , the value of inductive reactance will be :
 (A) $R/4$ (B) $R/2$
 (C) R (D) Cannot be found with the given data

51. An electro magnetic wave of wavelength 300 metre can be transmitted by a transmission centre. A condenser of capacity $2.5 \mu\text{F}$ is available. Calculate the inductance of the required coil for a resonant circuit. Use $\pi^2 = 10$.
52. In an AC circuit the potential differences across an inductance and resistance joined in series are respectively 16V and 20V. The total potential difference across the circuit is :
 (A) 20 V (B) 25.6 V (C) 31.9 V (D) 53.5 V
53. An AC voltage source $V = 200\sqrt{2} \sin 100 t$ is connected across a circuit containing an AC ammeter (it reads RMS value) and capacitor of capacity $1 \mu\text{F}$. The reading of ammeter is :
 (A) 10 mA (B) 20 mA (C) 40 mA (D) 80 mA
54. A transformer has 50 turns in the primary and 100 in the secondary. If the primary is connected to a 220 V DC supply, what will be the voltage across the secondary ?
55. r.m.s. value of current $i = 3 + 4 \sin (\omega t + \pi/3)$ is :
 (A) 5 A (B) $\sqrt{17}$ A (C) $\frac{5}{\sqrt{2}}$ A (D) $\frac{7}{\sqrt{2}}$ A
56. A coil of inductance 5.0 mH and negligible resistance is connected to an alternating voltage $V = 10 \sin (100 t)$. The peak current in the circuit will be :
 (A) 2 amp (B) 1 amp (C) 10 amp (D) 20 amp
57. An AC voltage is given by

$$E = E_0 \sin \frac{2\pi t}{T}$$
 Then the mean value of voltage calculated over time interval of $T/2$ seconds :
 (A) is always zero (B) is never zero (C) is $(2E_0/\pi)$ always (D) may be zero
58. Average value of A.C. current in a half time period may be :
 (A) positive (B) negative (C) zero (D) None of these
59. If the frequency of the source e.m.f. in an AC circuit is n , the power varies with a frequency :
 (A) n (B) $2n$ (C) $n/2$ (D) zero
60. An AC voltage of $V = 220\sqrt{2} \sin \left(2\pi 50t + \frac{\pi}{2} \right)$ is applied across a DC voltmeter, its reading will be :
 (A) $220\sqrt{2}$ V (B) $\sqrt{2}$ V (C) 220 V (D) Zero
61. In a transformer ratio of secondary turns (N_2) and primary turns (N_1) i.e. $\frac{N_2}{N_1} = 4$. If the voltage applied in primary is 200 V, 50 Hz. Find (a) voltage induced in secondary (b) If current in primary is 1A, find the current in secondary if the transformer is (i) ideal and (ii) 80% efficient and there is no air loss.
62. A power (step up) transformer with an 1 : 8 turn ratio has 60 Hz, 120 V across the primary; the load in the secondary is $10^4 \Omega$. The current in the secondary is :
 (A) 96 A (B) 0.96 A (C) 9.6 A (D) 96 mA
63. A series LCR circuit with $L = 0.125/\pi$ H, $C = 500/\pi$ nF, $R = 23 \Omega$ is connected to a 230 V variable frequency supply.
 (a) What is the source frequency for which current amplitude is maximum ? Obtain this maximum value.
 (b) What is the source frequency for which average power absorbed by the circuit is maximum ? Obtain the value for this maximum power.
 (c) For what reactance of the circuit, the power transferred to the circuit is half the power at resonance ? What is the current amplitude at this reactance ?
 (d) If ω is the angular frequency at which the power consumed in the circuit is half the power at resonance, write an expression for ω
 (e) What is the Q -factor of the given circuit.

64. (i) The overall efficiency of a transformer is 90%. The transformer is rated for an output of 9000 watt. The primary voltage is 1000 volt. The ratio of turns in the primary to the secondary coil is 5 : 1. The iron losses at full load are 700 watt. The primary coil has a resistance of 1 ohm.
 (A) 1000 volt (B) 5000 volt (C) 200 volt (D) zero volt
- (ii) In the above, the current in the primary coil is :
 (A) 9 amp (B) 10 amp (C) 1 amp (D) 4.5 amp
- (iii) In the above, the copper loss in the primary coil is :
 (A) 100 watt (B) 700 watt (C) 200 watt (D) 1000 watt
- (iv) In the above, the copper loss in the secondary coil is :
 (A) 100 watt (B) 700 watt (C) 200 watt (D) 1000 watt
- (v) In the above, current in the secondary coil is :
 (A) 45 amp (B) 46 amp (C) 10 amp (D) 50 amp
- (vi) In the above, the resistance of the secondary coil is approximately :
 (A) 0.01 Ω (B) 0.1 Ω (C) 0.2 Ω (D) 0.4 Ω
65. A transformer is used to light a 140 watt, 24 volt lamp from 240 V AC mains. The current in the main cable is 0.7 amp. The efficiency of the transformer is :
 (A) 48% (B) 63.8% (C) 83.3% (D) 90%
66. In a step-up transformer the voltage in the primary is 220 V and the current is 5 A. The secondary voltage is found to be 22000 V. The current in the secondary (neglect losses) is :
 (A) 5 A (B) 50 A (C) 500 A (D) 0.05 A
67. The core of a transformer is laminated to reduce :
 (A) Eddy current loss (B) Hysteresis loss (C) Copper loss (D) Magnetic loss
68. A capacitor is a perfect insulator for :
 (A) Direct current (B) Alternating current
 (C) Direct as well as alternating current (D) None of these
69. A choke coil should have :
 (A) high inductance and high resistance (B) low inductance and low resistance
 (C) high inductance and low resistance (D) low inductance and high resistance
70. A choke coil is preferred to a rheostat in AC circuit as :
 (A) it consumes almost zero power (B) it increases current
 (C) it increases power (D) it increases voltage
71. With increase in frequency of an AC supply, the inductive reactance :
 (A) decreases (B) increases directly proportional to frequency
 (C) increases as square of frequency (D) decreases inversely with frequency
72. A circuit containing a 0.1 H inductor and a 500 μ F capacitor in series is connected to a 230 volt, $100/\pi$ Hz supply. The resistance of the circuit is negligible. (a) Obtain the current amplitude and rms values. (b) Obtain the rms value of potential drops across each element. (c) What is the average power transferred to the inductor ? (d) What is the average power transferred to the capacitor ? (e) What is the total average power absorbed by the circuit ? ['Average' implies average over one cycle.]
73. With increase in frequency of an AC supply, the capacitive reactance :
 (A) varies inversely with frequency (B) varies directly with frequency
 (C) varies directly as square of frequency (D) remains constant

74. An AC ammeter is used to measure current in a circuit. When a given direct constant current passes through the circuit, the AC ammeter reads 3 ampere. When another alternating current passes through the circuit, the AC ammeter reads 4 ampere. Then the reading of this ammeter if DC and AC flow through the circuit simultaneously, is :
 (A) 3 A (B) 4 A (C) 7 A (D) 5 A
75. In an AC circuit consisting of resistance R and inductance L , the voltage across R is 60 volt and that across L is 80 volt. The total voltage across the combination is :
 (A) 140 V (B) 20 V (C) 100 V (D) 70 V
76. An LCR circuit has $L = 10 \text{ mH}$, $R = 150 \Omega$ and $C = 1 \mu\text{F}$ connected in series to a source of $150\sqrt{2} \cos \omega t$ volt. At a frequency that is 50% of the resonant frequency, calculate :
 (a) the net reactance of the circuit.
 (b) the current amplitude and the average power dissipated per cycle.
77. A 2000 Hz, 20 volt source is connected to a resistance of 20 ohm, an inductance of $0.125/\pi \text{ H}$ and a capacitance of $500/\pi \text{ nF}$ all in series. Calculate the time in which the resistance (thermal capacity = $100 \text{ joule/}^\circ\text{C}$) will get heated by 10°C .
78. Find the rms value for the saw-tooth voltage of peak value V_0 from $t = 0$ to $t = 2T$ as shown in figure :



* * * * *

ANSWER KEY

1. (A) 2. (A) 3. (D) 4. (C)(D) 5. $[220\sqrt{2} \text{ V}, 2.5 \text{ ms}]$ 6. (B) 7. $[\frac{I_0}{e} \sqrt{(e^2 - 1)/2}]$
8. (C) 9. (A) 10. $[I_{\text{eff}} = [a^2 + \frac{1}{2}b^2]^{1/2}]$ 11. [0] 12. (B) 13. (A) 14. $[12\sqrt{2} \text{ volts}]$
15. (B) 16. (C) 17. (B) 18. (C) 19. (B) 20. (B) 21. (A) 22. (B) 23. (All)
24. [0.72 W] 25. [10 W, 5 W] 26. (D) 27. (B) 28. (A) 29. [3.0 kV] 30. (B) 31. (A)(B)(D)
32. (C) 33. $[\frac{2.2\sqrt{3}}{\pi} = 1.2 \text{ H}]$ 34. [(a) 0.1 A (b) 50 V, 30 V, 10 V (Note that the sum of the rms potential differences across the three elements is greater than the rms voltage of the source.)] 35. [25 mJ, 5 mJ] 36. (B) 37. (C)
38. (D) 39. [(i) $\frac{125}{33} \mu\text{F}$ (ii) 2.4 H (iii) 720Ω (iv) It will be more economical to use inductance or capacitance in series with the lamp to run it as it. It consumes no power while there would be dissipation of power when resistance is inserted in series with the lamp.] 40. (A) 41. (D) 42. $[Z = 50\sqrt{2} \text{ ohm}, V_C = 500\sqrt{2} \text{ volt and } V_L = 600\sqrt{2} \text{ volt}]$
43. [1 amp, $\cos \phi = \frac{25}{\sqrt{577}}$] 44. $[\sqrt{\frac{21}{22}} \text{ H (1 H)}, 100 \Omega]$ 45. [125 Ω , 288 J] 46. $[4 \mu\text{F}, R = \frac{141.4}{5} \Omega]$
47. [(a) $\frac{250}{3\pi} \text{ Hz}$ (b) 2 mA] 48. [1.5 A] 49. (D) 50. (C) 51. $[1 \times 10^{-8} \text{ Henry}]$ 52. (B) 53. (B)
54. [Zero] 55. (B) 56. (D) 57. (D) 58. (A)(B)(C) 59. (B) 60. (D)
61. [(a) 800 V (b) (i) 0.25 A (ii) 0.2 A] 62. (D)
63. [(a) 2000 Hz, $10\sqrt{2} \text{ A}$ (b) 2000 Hz, 2300 watt (c) 23Ω , 10 A (d) $\frac{0.125}{\pi} \omega - \frac{1 \times 10^9}{\omega \frac{500}{\pi}} = \pm 23$ (e) 500/23]
64. (i) (C) (ii) (B) (iii) (A) (iv) (C) (v) (B) (vi) (B) 65. (C) 66. (D) 67. (A)
68. (A) 69. (C) 70. (A) 71. (B) 72. [(a) $23\sqrt{2} \text{ A}$, 23 A (b) 460 volt, 230 volt (c) zero (d) zero (e) zero]
73. (A) 74. (D) 75. (C) 76. [(a) 150Ω (b) 1 amp, 75 watt] 77. [50 sec] 78. $[\frac{V_0}{\sqrt{3}}]$